KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE HEALTH SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRONICS

DATE: 28th September, 2022	**TIME: 3 HOURS (9:00 A.M – 12 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

SECTION 1 - MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The number of PN junctions in a thyristor are
   a) 1
   b) 2
   c) 3
   d) 4
2. A thyristor can be termed as a
   a) DC switch
   b) AC switch
   c) Both A and B
   d) Square ware switch
3. The following are statements regarding a TRAC.
   i. It is unidirectional
   ii. It is equivalent to two SCRs connected to ANTI parallel when in operation
   iii. It can be turned on by applying the positive and negative voltage to the gate with respect to the MTI terminal.

   Which statement(s) is correct?
   a) i
   b) i and ii
   c) ii only
   d) ii and iii only
4. Given that the holding current level of an SCR is I_H, latching current level of an SCR is I_L and its anode current is I_A, then an SCR is turned off when
   a) $I_A > I_H$
   b) $I_A < I_H$
   c) $I_A = I_l$
   d) $I_A < I_l$
5. As gate current increases forward break over voltage will
   a) Remain the same
   b) Decrease
   c) Increase
   d) Increase and then decrease
6. Linear Variable Differential Transformer (LVDT) is a
   a) Capacitive transducer
   b) Resistive transducer
   c) Inductive transducer
   d) None of the above
7. Piezoelectric crystals are used for measurement of _____ charges.
   a) Static
   b) Dynamic
   c) Static and dynamic
   d) All of the above

8. A piezoelectric crystal produces an E.M.F. when
   a) External mechanical force is applied
   b) External magnetic field is applied
   c) Radiant energy stimulates
   d) When the junction of the two crystal is treated
9. Which one of the following instruments can be used for the measurement of temperature above 1500^0C?
   a) Mercury thermometer
   b) Gas thermometer
   c) Thermal electric pyrometer
   d) All of the above
10. Thermistors are made of
   a) Pure metals
   b) Pure insulators
   c) Sintered mixtures of metallic oxides
   d) Pure semiconductors
11. The main feature of a large signal amplifier circuits is
   a) Power efficiency
   b) Distortion introduced by the circuit
   c) Heat dissipated by the circuit
   d) All of the above
12. Class _____ amplifiers are normally operated in a push pull configuration in order to produce an output that is a replica of the input.
   a) A
   b) B
   c) C
   d) AB
13. The Q-point is at cut off for class _____ operation.
   a) A
   b) B
   c) C
   d) AB
14. How many transistors must be used in a class B amplifier to obtain an output for the full cycle of the input signal?
   a) 0
   b) 1
   c) 2
   d) 3
15. By how much does the output signal vary for a class A power amplifier?
   a) 360°
   b) 180°
   c) Between 180° and 360°
   d) Less than 180°

SECTION 1 - MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The number of PN junctions in a thyristor are
   a) 1
   b) 2
   c) 3
   d) 4
2. A thyristor can be termed as a
   a) DC switch
   b) AC switch
   c) Both A and B
   d) Square ware switch
3. The following are statements regarding a TRAC.
   i. It is unidirectional
   ii. It is equivalent to two SCRs connected to ANTI parallel when in operation
   iii. It can be turned on by applying the positive and negative voltage to the gate with respect to the MTI terminal.

   Which statement(s) is correct?
   a) i
   b) i and ii
   c) ii only
   d) ii and iii only
4. Given that the holding current level of an SCR is I_H, latching current level of an SCR is I_L and its anode current is I_A, then an SCR is turned off when
   a) $I_A > I_H$
   b) $I_A < I_H$
   c) $I_A = I_l$
   d) $I_A < I_l$
5. As gate current increases forward break over voltage will
   a) Remain the same
   b) Decrease
   c) Increase
   d) Increase and then decrease
6. Linear Variable Differential Transformer (LVDT) is a
   a) Capacitive transducer
   b) Resistive transducer
   c) Inductive transducer
   d) None of the above
7. Piezoelectric crystals are used for measurement of _____ charges.
   a) Static
   b) Dynamic
   c) Static and dynamic
   d) All of the above

8. A piezoelectric crystal produces an E.M.F. when
   a) External mechanical force is applied
   b) External magnetic field is applied
   c) Radiant energy stimulates
   d) When the junction of the two crystal is treated
9. Which one of the following instruments can be used for the measurement of temperature above 1500^0C?
   a) Mercury thermometer
   b) Gas thermometer
   c) Thermal electric pyrometer
   d) All of the above
10. Thermistors are made of
   a) Pure metals
   b) Pure insulators
   c) Sintered mixtures of metallic oxides
   d) Pure semiconductors
11. The main feature of a large signal amplifier circuits is
   a) Power efficiency
   b) Distortion introduced by the circuit
   c) Heat dissipated by the circuit
   d) All of the above
12. Class _____ amplifiers are normally operated in a push pull configuration in order to produce an output that is a replica of the input.
   a) A
   b) B
   c) C
   d) AB
13. The Q-point is at cut off for class _____ operation.
   a) A
   b) B
   c) C
   d) AB
14. How many transistors must be used in a class B amplifier to obtain an output for the full cycle of the input signal?
   a) 0
   b) 1
   c) 2
   d) 3
15. By how much does the output signal vary for a class A power amplifier?
   a) 360°
   b) 180°
   c) Between 180° and 360°
   d) Less than 180°

16. A differential amplifier has an open-loop voltage gain of 100. The input signals are 2.5V and 2.4 V. What is the output voltage of the amplifier?
    a) -10V
    b) 1mV
    c) 10V
    d) 1KV
17. Which one of the following statements relating to an operational amplifier is true?
    a) It has a high-open loop voltage gain of low frequency and a low output impedance and low output impedance
    b) It has a high-open loop gain at low frequency, a high input impedance and low output impedance
    c) It has a low open-loop voltage gain at low frequency, a high input impedance and low output impedance
    d) It has a high open-loop voltage gain at low frequency, a low input impedance and high output impedance
18. A differential amplifier
    a) Is a part of an operational amplifier
    b) Has one input and one output
    c) It has two outputs
    d) Both A and B
19. The common mode gain of an operational amplifier is
    a) Very high
    b) Very low
    c) Always unity
    d) Unpredictable
20. Negative feedback
    a) Increases the input and output impedances
    b) Increases the input impedance and bandwidth
    c) Decreases output impedance and bandwidth
    d) Does not affect impedance or bandwidth
21. Oscillator converts
    a) A.C power into D.C. power
    b) D.C power into A.C. power
    c) Mechanical power into A.C power
    d) None of the above
22. In an LC transistor circuit, the active device is
    a) LC tank circuit
    b) Braising circuit
    c) Transistor
    d) None of the above
23. In an LC circuit, when the capacitor energy is maximum, the inductor energy is
    a) Minimum
    b) Maximum
    c) Half way in between maximum and minimum
    d) The quarter at maximum

24. An oscillator employs _____ feedback.
    a) Positive
    b) Negative
    c) Both positive and negative
    d) Open loop feedback
25. Hartley Oscillator is commonly used in
    a) Radio receivers
    b) Radio transmitter
    c) TV receivers
    d) Clocking of micro processors
26. In a phase shift oscillator, the frequency determining elements are
    a) L and C
    b) R, L and C
    c) R and C
    d) L and R
27. In crystal oscillator, frequency is stable due to _____ of the crystal.
    a) Rigidity
    b) Vibrations
    c) Low Q
    d) High Q
28. An oscillator differs from an amplifier because it
    a) Has more gain
    b) Requires no input signal
    c) Required no DC supply
    d) Always has the same input
29. In a colpitts' crystal oscillator, feedback is obtained
    a) By magnetic induction
    b) By a tickle coil
    c) Form center of split capacitors
    d) From center of split inductors
30. _____ is a fixed frequency oscillator.
    a) Phase-shift oscillator
    b) Hartley Oscillator
    c) Culpitts' Oscillator
    d) Crystal Oscillator
31. When a transistor is driven to saturation, ideally the output voltage is
    a) V_{CC}
    b) 0
    c) $V_{CC}/2$
    d) $2V_{CC}$
32. When a transistor is driven to cut off, ideally the output voltage is
    a) V_{CC}
    b) 0
    c) $2V_{CC}$
    d) $V_{CC}/2$

33. Which multivibrator is a square wave oscillator?
    a) Monostable
    b) Astable
    c) Bistable
    d) Tristable
34. A bistable multivibrator has
    a) Two stable state
    b) One stable state
    c) No stable state
    d) Three stable states
35. A multivibrator in which the circuit is stable in either state and can be flipped from one state to the other by an external trigger pulse is termed as
    a) Astable
    b) Monostable
    c) Bistable
    d) Tristable
36. The binary number 10101 is equivalent to decimal number
    a) 19
    b) 12
    c) 27
    d) 21
37. The universal gate is _____ amongst the following.
    a) NAND gate
    b) OR gate
    c) AND gate
    d) NOT gate
38. Which one of the following gates is an inverter?
    a) NOT gate
    b) OR gate
    c) AND gate
    d) None of the above
39. In Boolean Algebra, the bar sign (-) indicates
    a) OR operation
    b) AND operation
    c) NOT operation
    d) EXOR operation
40. Decimal 10 is equal to binary number
    a) 1110
    b) 1010
    c) 1001
    d) 1000

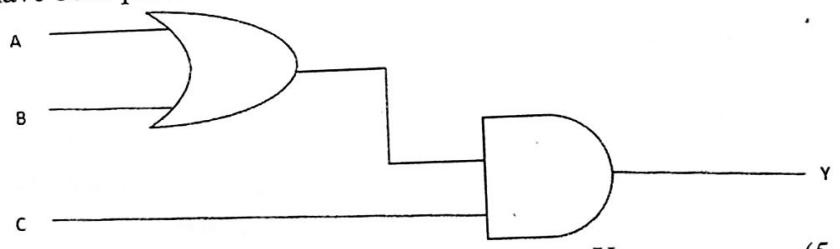
SECTION 2 – SHORT ANSWER QUESTIONS (40 MARKS)

41. (a) What is a transducer? (2 marks)
    (b) State three factors considered when selecting a transducer. (3 marks)

42. Outline the differences between voltage and power amplifiers. (5 marks)
43. List five ideal characteristics of an operational amplifiers. (5 marks)
44. (a) State and explain three conditions that must be fulfilled for a tank circuit to produce undamped oscillations. (3 marks)
(b) Convert 12_{10} to octal number system. (2 marks)
45. (a) State any four advantages of crystal oscillators. (4 marks)
(b) State one disadvantage of a crystal oscillator. (1 marks)
46. A transistor is used as a switch. If $V_{CC} = 10V$, $R_C = 1K\Omega$. Determine the value of V_{CE} at cutoff and I_C at saturation. (5 marks)
47. (a) What is a multivibrator. (2 marks)
(b) Draw a 3-input EX-OR gate. (3 marks)
48. Draw the schematic symbol for the following gates: (5 marks)
i. AND
ii. NOT
iii. OR
iv. NAND
v. NOR

SECTION 3 – LONG ANSWER QUESTION (20 MARKS)

49. (a) With the aid of a well labelled diagram, explain the working principle of a UJT relaxation oscillator. (10 marks)
(b) You have been provided with the following digital circuit.



(i) Work out the Boolean expression for output Y. (5 marks)
(ii) Develop the truth table for the output Y using the three inputs A, B and C. (5 marks)